

How Far Can We Push Flow Analysis to Identify Encrypted Anonymity Network Traffic?

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Abstract— Anonymity networks provide privacy to the users by relaying their data to multiple destinations in order to reach the final destination anonymously. Multilayer of encryption is used to protect the users' privacy from attacks or even from the operators of the stations. In this research, we showed how flow analysis could be used to identify encrypted anonymity network traffic under four scenarios: (i) Identifying anonymity networks compared to normal background traffic; (ii) Identifying the type of applications used on the anonymity networks; (iii) Identifying traffic flow behaviors of the anonymity network users; and (iv) Identifying / profiling the users on an anonymity network based on the traffic flow behavior. In order to study these, we employ a machine learning based flow analysis approach and explore how far we can push such an approach.

Keywords— *Dataset; Tor; I2P; JonDonym; Traffic Flow; Anonymity*

I. INTRODUCTION

Anonymity networks aim to provide the users with some level of privacy. These networks enable the users on the Internet to freely access websites or run applications on the Internet without revealing their identity to any site that observes the network. In addition, anonymity networks hide the users' identity from the final destination (web server). Some of the most used and known anonymity networks are Tor [1], JonDonym [2], and I2P [3].

Anonymity networks (services) have much in common in more than one aspect. They provide services to the users on the Internet while keeping their identity hidden. In addition, most of the time, anonymity services provide anonymity to the users by forwarding the users' traffic through multiple stations until the users' data reach to their destination. During this journey, the data are encrypted multiple times. This way, the users' data stay anonymous where each station during the journey knows only part of the information. Therefore, it becomes difficult to trace the users' data in these networks.

At the same time, these networks have differences between them. They differ in the design; this includes how the data is forwarded between the stations, the type of the encryption used, the way to manage the networks, the protocol to select the stations, the operators of the stations, and other details regarding the design. On the other hand, these networks have different goals under the design and different supported applications within the networks. For example, the anonymity networks could be designed specifically to browse the Internet websites anonymously. Others could be designed to be a private network

that users trade information internally. Such a network is not optimized to browse Internet web sites. The supported applications on these networks are different, such as browsing, file sharing, IRC, web hosting, etc. One type of difference is the operators of the networks; some anonymity networks count on volunteers to provide the service to the users. Others use companies to provide the service. In addition, the threat models of these networks are not the same based on their design. What could be a threat to one of these networks might not form any threat to the others.

Tor [1] [15], JonDoNym [2] and I2P [3] are the three well-known anonymity networks that we included on the proposed data set. Tor provides the users with anonymity by passing the traffic through three stations (nodes). These nodes run by volunteers shared their bandwidth with the anonymity network. The selection of these nodes is done by a protocol called path selection protocol. The users get the list of the currently available nodes on the networks from directory authorities servers. The path that the users use is not fixed, it continuously keeps changing. The user has the option to define the first or the last node on the path instead of counting on the path selection protocol to select the nodes.

JonDonym provides anonymity service to the users by passing the traffic through multiple stations (Mixes). The path (cascade) on the JonDonym network is fixed. The user can select the cascade that will relay the traffic but cannot change the mixes on the path. This is due to the difference in design of the anonymity networks. The mix on the JonDonym network multiplexes traffic from different users and sends it to the next mix on the cascade.

The Invisible Internet Project (I2P) is a packet switched network that uses multilayer encryption to provide anonymity to its users. The path (Tunnels) on the I2P network is unidirectional. The user builds an encrypted tunnel to the final destination to send messages using the created tunnel (outbound tunnel). The messages travel in one direction only; therefore, messages from destination to the sender must use another tunnel (inbound tunnel). Tunnels on the I2P networks are used for communication and management. By default, the users on the I2P network share bandwidth by participating on building I2P Tunnels. The user has the option to reduce or increase the amount of participation on building I2P Tunnels.

The connection between the user and any of these anonymity networks is not hidden. These networks aim to provide anonymity to the user, but they do not hide that the user is

connecting to such anonymity network. Therefore, blocking these anonymity networks is possible. Thus, Tor employed different obfuscation techniques (Pluggable Transports) to hide the connection to the Tor network. The implemented pluggable transports are Scramblesuit [9], Flashproxy [10], Obfs3 [11], Meek [12], Format-transforming encryption (FTE) [13]. Furthermore, JonDonym offers two options to resist the network blockage: TCP/IP forward and Skype tunnel. Both could be used to connect the user to the JonDonym network if the network is blocked. On the I2P network, the obfuscation options are considered but not implemented yet.

Anonymity networks have differences in the design, the applications they support, the main goal of the network and how the network resists censorship. In addition, the encryption employed in anonymity networks prevents using payload analysis to perform traffic analysis. On the other hand, flow analysis deals with statistical information extracted from the header of the traffic without the need to deal with the encryption. Therefore, in this paper, we explore the boundaries of flow analysis employment in anonymity network analysis.

The rest of this paper is organized as follows. Related work is reviewed in Section II. The Anon17 data set and the flow analysis approach used in this research are discussed in Section III. Section IV presents the use of flow analysis to identify anonymity networks. Section V presents the analysis of identifying the type of application on the anonymity network traffic. Whereas Section VI presents the analysis for identifying the traffic flows of users as well as user profiles. Finally, conclusions are drawn, and the future work is discussed in Section VII.

II. RELATED WORK

Flow analysis is applied in many researches to solve different types of problems. One of the applications of using flow analysis is encrypted traffic classification. Anonymity networks traffic is based on multiple layers of encryption. Therefore, flow analysis is studied to identify anonymity networks. Barker et al. [8] collected Tor data using a simulated environment. The goal of their research is to study the possibility to differentiate between encrypted traffic and Tor traffic. The data included HTTP and HTTPS traffic over the simulated Tor network and HTTPS traffic. NetAI was used to export flows from the captured traffic. The exported flows were analyzed using three machine learning algorithms namely, Random Forest, J4.8, and Adaboost. The result showed that it is possible to differentiate between Tor and encrypted traffic with more than 90% accuracy.

Machine learning algorithms are also used to identify the type of application running anonymously on an anonymity network. Alsabah et al. [17] used Naïve Bayes, Bayesian Networks, and Decision Tree algorithms to classify the application used by Tor's users. The applications are Browsing, Streaming and BitTorrent. Given that Tor traffic is encrypted, they used the circuit level and the cell level information to do the classification. The circuit level information included the circuit lifetime and the amount of data transferred by the circuit. The cell level information included the cells inter arrival time and their statistics. The classification included online and offline classification. The online classification used the cell level

information to classify the circuit while it is in use. The offline classification used both the cell and the circuit level information to classify the circuit. In terms of accuracy, the best result they achieved in the offline classification was 91%, whereas the best results achieved in the online classification was 97.8%.

In terms of the data set used on anonymity networks studies, researches in this field used simulation environments to generate the data or collected the data from the anonymity network. Bauer et al. [7] presented a Tor network emulation tool, namely ExperimentTor. The tool provides a test environment for the Tor researchers by modeling Tor routers, bandwidth, users, and applications. The ExperimentTor tool is available as standalone and a VMware image that has the tool installed and configured.

There are many other researches on Tor, JonDonym, and I2P anonymity networks [16] [17] [18] [19] where researchers used their own data to conduct their research. In this work, we also collect our own data on three anonymity tools, namely Tor, JonDonym, and I2P. Furthermore, we make this dataset, Anon17, publicly available for further research.

III. DATASET AND FLOW ANALYSIS

Flow analysis calculates statistical information extracted from the header of the packet to describe the communication between two parties. The following sections describe how the anonymity network data was collected and the flow exporter tool was used to generate the flows.

A. Anon17

There are many studies conducted on these anonymity networks. The researches include a wide range of aspects related to the anonymity field such as improving the design, performing attacks on the anonymity network, analyzing the users' behavior on the anonymity network, studying the performance and delay, revealing the users' identity, and many others.

In some of the anonymity researches, the used data are collected in a simulated environment. Others used real data collected by the researchers themselves. The most common issue that faces researchers in the anonymity field is that these anonymity networks provide anonymity to the users; thus, collecting the data and making it publicly available might affect the privacy of users of the anonymity tools. Therefore, the researches on the anonymity field count on data collected from a simulated environment or collected by the researchers themselves.

Anon17 [14] is the dataset we used in this work to identify anonymity networks. Anon17 is collected at the NIMS lab [6] between 2014-2017 in a real network environment. The dataset is labeled based on the information that we could extract from the anonymity networks while collecting the data. For example, in the Tor network, the IP addresses of the Tor nodes are available. Therefore, whenever we collect data related to a node on the Tor network, we use the IP address to label this traffic as "Tor". The same applies for all the labeling in our data as detailed below. We did not use any application classification tools to label our data. Instead, we used the ground-truth that we know to label all the traffic in Anon17. The port number was used only to label the background traffic collected by other research group. The background traffic was classified into

multiple applications / protocols to investigate the ability to distinguish between the anonymity traffic and different background application.

Anon17 contains data for three popular and well-known anonymity networks: Tor, JonDonym, and I2P. We provide these data and make it publicly available without affecting the privacy of the users. To this end, the IP addresses of the users have been removed. The payload information only used for statistical measurement and then are removed. This is because we aim to provide a publicly available anonymity dataset that could be used to study the aforementioned anonymity tools. The dataset includes several applications used on these anonymity networks and also includes several obfuscation techniques that are used on some of these networks. Therefore, the dataset could be used for multiple types of researches.

The following details the content of Anon17 used in this paper and our previous works:

1) *Tor*

The Tor dataset contains Tor traffic. The traffic includes the circuit establishment and the user activities on the Tor network such as browsing the Internet websites. The Tor traffic does not include any obfuscated traffic. The data collected from multiple machines in NIMS labs while using Tor to browse Internet web pages.

2) *TorApp*

The TorApp dataset contains flows for three machines (computers) running three applications on the Tor network (Browsing, Video streaming, and File sharing). Therefore, there are three classes on the TorApp dataset (Browsing, streaming, and BitTorrent). The Browsing class contains connection flows between a user and an entry node on the Tor network when the user is using Tor to browse different Internet websites. The Streaming class is the connection flows between the user and the entry node when the user is watching videos on Tor. The last class, the BitTorrent class contains flows between the user and the first node when the user is using Torrent files on the Tor network

3) *TorPT*

The TorPT dataset contains flows for Tor pluggable transports. The TorPT has five classes: Obfs3, Meek, Flashproxy, FTE, and Scramblesuit. The TorPT is collected by connecting to the pluggable transports from the NIMS lab and capture the traffic. The Obfs3 is collected from two different Obfs3 bridges. The Scramblesuit traffic is collected by connecting to 22 different Scramblesuit servers. This is to ensure that we include the effect of changing the flow behavior that Scramblesuit pluggable transport aims to achieve.

4) *I2PApp80BW*

These traffic flows are collected while running three applications on the I2P network. The applications (classes) are Eepsites (the websites browsing on the I2P network), jIRCii (Internet Relay Chat (IRC) plugin on the I2P network), and I2Psnark (the file sharing plugin on the I2P network). The bandwidth sharing on the I2P client is set to default which is 80% sharing rate of the user bandwidth. In this dataset, each class contains the application flows in addition to the management traffic flows. For example, the Eepsites flows

contain flows for the Eepsites Tunnels in addition to the Tunnels used for the management of the I2P network and tunnels used to share bandwidth such as the Exploratory and the Participating Tunnels.

5) *I2PApp0BW*

This dataset is similar to the I2PApp80BW; the difference is that the amount of shared bandwidth is set to 0%. This will reduce the amount of management traffic flows on each class.

6) *I2PUsers*

This dataset contains the traffic flows for three users on the I2P network. The classes are named PC1, PC2, and PC3. The dataset I2PUsers is the same dataset used in I2PApp80BW. The difference is that the data is classified differently. Instead of labeling the dataset based on the application; the dataset is labeled based on the user traffic. Therefore, any class on this dataset will contain the three applications used on the I2PApp80BW dataset. For example, the PC1 flows contain flows for the machine of the first users when the users used Eepsites, jIRCii, and I2Psnark on the I2P network.

7) *I2PApp*

This dataset contains traffic flows for the same three applications used in I2PApp80BW. The difference is that this dataset contains separate classes for the management tunnels. The total number of classes on this dataset is five: Eepsites, jIRCii, I2Psnark, Exploratory Tunnels, and Participating Tunnels. Therefore, the application tunnels do not contain any management tunnels flows.

8) *JonDonym*

The JonDonym dataset contains traffic flows for the JonDonym network as well as traffic flows for the whole free mixes in the JonDonym network [14].

B. *Flow Exporter*

We used Tranalyzer [4] to extract the flows from the PCAP files we captured in the NIMS lab. Tranalyzer has 92 features based on the plug-ins used such as Number of bytes sent, Number of bytes received, Statistics about the Inter-arrival time, and Number of connection etc. Some of the unrelated features are removed from the dataset such as the ICMP features and VLAN features because they do not provide useful information for our purposes. IP addresses and payloads of the packets are also removed from the dataset to protect the privacy of the users. In the dataset, the values of some features might have zeros. For example, the I2P network works on both TCP and UDP. Therefore, if the I2P data set contains UDP connections then all the TCP related features will have zero values.

Tranalyzer has both the flow exporter and the flow collector integrated into the Tranalyzer. Figure 1. Shows the setup for collecting the data in Anon17.

The data is formatted into arff file format used in the open source data mining software tool, Weka [5]. Table I summarizes the features included in Anon17 dataset.

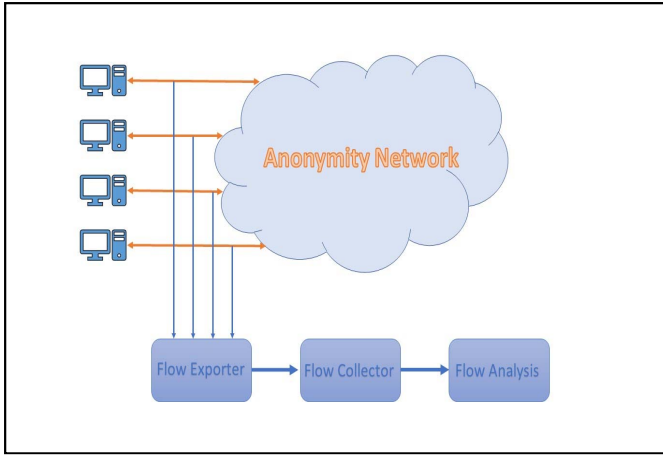


Fig. 1. Anon17 Flow Analysis.

TABLE I. ANON17 DATASET FEATURES

Features	Description
dir time_first, time_last, duration	Flow direction, time, and duration of the flow
numPktsSnt numPktsRcvd numBytesSnt numBytesRcvd minPktSz maxPktSz avePktSize pktps bytps pktAsm bytAsm	Counting of Packets and Bytes
ip_mindIPID ip_maxdIPID ip_minTTL ip_maxTTL ip_TTL_Chg ip_TOS ip_flags ip_Opt ip_OptCnt	The IP Header related features such as TOS, TTL etc.
tcp_PSeqCnt tcp_SeqSntBytes tcp_SeqFaultCnt tcp_PAckCnt tcp_FlwLssAckRcvdBytes tcp_AckFaultCnt tcp_InitWinSz tcp_AveWinSz tcp_MinWinSz tcp_MaxWinSz tcp_WinSzDwnCnt tcp_WinSzUpCnt tcp_WinSzChgDirCnt tcp_AggrFlags tcp_AggrAnomaly tcp_AggrOptions tcp_MSS tcp_WS tcp_OptCnt tcp_S- SA/SA-A_Trip tcp_S-SA-A/A-A_RTT tcp_RTTAckTripMin tcp_RTTAckTripMax tcp_RTTAckTripAve tcpStates	The TCP Header related features such as Window size, sequence number etc.
connSrc connDst connSrc<->Dst	Counting of number of connections between source and destination/ source to different destinations.
min_pl max_pl mean_pl low_quartile_pl median_pl upp_quartile_pl iqd_pl mode_pl range_pl std_pl stdrob_pl skew_pl exc_pl	Packet length statistics
min_iat max_iat mean_iat low_quartile_iat median_iat upp_quartile_iat iqd_iat mode_iat range_iat std_iat stdrob_iat skew_iat exc_iat nfp_pl_iat ps_iat_histo	Inter arrival time statistics
TrafficType	The classes

IV. IDENTIFYING ANONYMITY NETWORKS

In this section, the flow analysis is used to identify an anonymity network among different background traffic. The selected anonymity network to be studied using flow analysis in this section is JonDonym. The JonDonym data employed in this analysis is the JonDonym part from the Anon17 dataset. The data of JonDonym are collected from three machines at NIMS lab by connecting to all the free cascades on the JonDonym network. For a background traffic, LBNL/ICSI [20] data set are employed as the background traffic. It contains network traces collected from more than 100 hours of activities for several thousands of hosts. The data size is 11 GB. The data are publicly available in a PCAP form. The data are distributed over several small PCAP files. For this analysis, a total of 211,370 flows are extracted from around 1.5 GB. Table II shows the result of the JonDonym traffic with the background traffic. The results show that JonDonym flows can be distinguished from background traces with a high accuracy level (99.99%).

TABLE II. RESULTS FOR JONDONYM FLOW ANALYSIS.

	TP Rate	FP Rate	Precision	F-Measure
JonDoNym	0.997	0	1	0.998
BackGround	1	0.003	1	1
LBNL/ICSI				
Accuracy	99.99%			

The next part of the analysis is performed by labeling the background data to application/protocol names based on the port number. The background data contains a vast number of applications and protocols. Some of the application or the protocols appear just a few times while others have a high number of appearances in the data. Therefore, the top appeared applications on the data are labeled with their application name, the rest are labeled as others. Thus, instead of having one class, namely background, the data will have 12 classes namely, HTTP, HTTPS, IMAPS, SNMP, NetBIOS-SSN, DNS, POP3, LPD, EPMAP, SMTP, SSH, and other. Table III shows the results of the applications and JonDonym analysis. The results dropped by 2% compared to the previous analysis where all the background traffic flows were grouped into one class.

V. IDENTIFYING THE OBFUSCATED ANONYMITY NETWORK TRAFFIC

The flow analysis could be used to explore the behavior of the anonymity network when anti-censorship techniques are used on these networks. Tor pluggable transports systems work to provide access to the Tor network in adversarial (censorship) environments. Most of the pluggable transports tools concentrate on hiding the content of the packets in a way that makes it hard for the adversaries when using deep packet inspection (DPI) to detect the connection to the bridges. But DPI is not the only method used to detect Tor traffic. The active probing and the flow analysis are some of the other popular methods used to detect Tor traffic. In our previous analysis [21], we demonstrated the resistance of Tor pluggable transports

against network traffic flow analysis techniques. Table IV shows the result of the flow analysis of Tor pluggable transports with different background traffic from our previous works. The pluggable transport traffic includes FTE, Scramblesuit, Meek, Flashproxy, and Obfs3. The pluggable transports related traffic could be found in the Anon17 datasets, under TorPT. The background traffic includes HTTP, HTTPS, SSH, BitTorrent, and Encrypted BitTorrent. In this case, the results show 97%.

TABLE III. RESULTS OF APPLICATIONS AND JONDONYM FLOW ANALYSIS

	TP Rate	FP Rate	Precision	F-Measure
HTTP	0.986	0.013	0.981	0.984
HTTPS	0.897	0.004	0.919	0.908
IMAPS	0.88	0.001	0.900	0.888
SNMP	0.998	0.000	0.996	0.997
NETBIOS-SSN	0.974	0.002	0.971	0.972
DNS	0.997	0.000	0.998	0.998
POP3	0.982	0.000	0.969	0.975
LPD	0.998	0.000	0.998	0.998
EPMAP	0.993	0.000	0.988	0.990
SMTP	0.948	0.000	0.961	0.955
SSH	0.455	0.000	0.613	0.522
OTHER	0.973	0.006	0.976	0.974
JonDonym	0.999	0.000	1.000	1.000
Accuracy	97.99%			

TABLE IV. APPLICATIONS ANALYSIS ON THE TOR NETWORK

	Classifier	Accuracy	F-Measure		
			Browsing	Streaming	BitTorrent
TRANALYZER2	Bayes net	99.2%	0.99	0.99	1
	Naïve Bayes	93.3%	0.98	0.90	0.93
	C4.5	97.2%	0.96	0.98	0.98
	Random Forest	98.8%	0.99	0.98	0.99
TCPTRACE	Bayes net	97.7%	0.98	0.97	0.99
	Naïve Bayes	92.2%	0.96	0.89	0.92
	C4.5	96.1%	0.96	0.95	0.98
	Random Forest	97.7%	0.99	0.96	0.98

VI. IDENTIFYING APPLICATIONS ON ANONYMITY NETWORKS

The flow analysis could be used to identify the applications on the anonymity networks. In our previous works, we studied the possibility to identify the application on two well-known anonymity networks (Tor [1] and I2P [3]). The following sections show the finding on identifying applications on anonymity networks.

A. Applications Classification on the Tor Network

Browsing, Video streaming, and BitTorrent are the three applications we studied in our previous work [21]. We set up

three computers to connect to our configured node on the Tor network and collected the flow of these three applications. The flow analysis is then employed to identify the type of application running on the Tor network. The flows are extracted using both Tranalyzer and Tcptrace. Four machine learning algorithms namely C4.5, Naïve Bayes, Random Forest, and Bayesian Network are used in the analysis of these three applications. Table IV shows the results of the four classifying algorithms for the two employed flow exporters. The accuracy is between 92% - 99% based on the machine learning algorithm and the flow exporter used. Table V shows the results of the Tranalyzer flow analysis against different background traffic. The flows of the three applications on Tor are included in Anon17 and called TorApp.

TABLE V. RESULTS FOR THE FLOW ANALYSIS OF TOR PLUGGABLE TRANSPORTS AGAINST DIFFERENT BACKGROUND TRAFFIC

	Class	TP Rate %	FP Rate %	Pre- cision %	F- Measure %
Background Traffic	HTTP	99	0.1	99	99
	HTTPS	94	0	95	95
	SSH	99	0	99	99
	BT	94	2.5	84	89
	BTecr	89	0.9	96	92
Pluggable Transports Traffic	FTE	99	0	99	99
	Scramble suit	98	0.1	92	95
	Meek	99	0	99	99
	Flash proxy	99	0.1	99	99
	Obfs3	99	0	99	99
Overall Correctly Classified Instances	97%				

B. Bandwidth and Applications Analysis on the I2P Network.

I2P network uses unidirectional tunnels for the outgoing and incoming traffic. The default setting on the I2P network enables bandwidth participations. This means that users on the I2P networks share their bandwidth to build tunnels that could be used by other I2P's users. The default setting is 80% bandwidth participating. In our previous work [22], we studied the effect of sharing bandwidth on the anonymity of I2P network using the flow analysis. We set up three machines to connect to the I2P network and collect data. Browsing, File downloading, and IRC are the three applications each machine run while collecting the data. The data is included in Anon17 and called I2P80BW. The bandwidth participation is then disabled, and the three-applications data collection is repeated. In Anon17, this part is called I2P0BW. The accuracy when using the flow analysis for the applications classification on the I2P networks shows low accuracy when the bandwidth participation is high. On the other hand, when the bandwidth participation is disabled, then the accuracy improved. The labeling in the I2P80BW and I2P0BW ignores that the tunnels on the I2P are used for data and also for managing the network. Therefore, the flows for any of the three applications are not pure application flows but also includes flows that belong to the management of the network. Therefore,

another experiment is conducted to collect data from the same three machines and for the same three applications where the management tunnels are separated from the data. This part of the data is called I2Papp. It contains five classes, three for the applications and two for the management part as explained in our previous work [22].

C. User Profiling on the I2P network

The same data used in section B is used to study flow analysis for profiling users on the I2P network. Instead of labeling the data based on the applications, the data is labeled based on the machine. In this set of experiments, each machine represents a different user. The I2PUsers in Anon17 represents the data for user profiling we used explore how far user profiling can be pushed on the I2P network. I2PUsers contains three classes, one for each machine. Table VI shows the results when using the flow analysis for both applications classification and users profiling when the bandwidth participating is disabled.

TABLE VI. APPLICATIONS CLASSIFICATION AND USERS PROFILING ON THE I2P NETWORKS

	Number of Instances	Accuracy (%)
Traffic Profiling	195,081	73.7
Traffic Profiling – TCP Only	40,075	65.6
Traffic Profiling – UDP Only	155,006	75.7
User Profiling	195,081	66.7
User Profiling – TCP Only	40,075	81.7
User Profiling – UDP Only	155,006	63.2

VII. CONCLUSION AND FUTURE WORK

In this paper, we showed the potential of using the flow analysis and its applications on the identification of the anonymity networks. The flow analysis has high accuracy in identification of the encrypted anonymity networks. We employed the flow analysis in three main categories: (i) identifying the anonymity networks and the ability to differentiate any encrypted traffic from encrypted anonymity networks traffic; (ii) identifying the obfuscated anonymity network traffic within the encrypted traffic; and (iii) applications profiling using the flow analysis. In our experiments, the accuracy varies based on the applications, the anonymity network, and the user's configuration when using the anonymity network. Furthermore, Anon17 is presented in this paper and made publicly available. Anon17 contains the traffic samples for three anonymity networks: Tor, JonDonym, and I2P. For future work, we will expand Anon17 to include additional applications running on these anonymity networks.

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